Adding USGS channel measurements to RiverShapeML

More gage sites, better-constrained targets, and a re-predicted national product with prediction intervals

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Summary

RiverShapeML (Modaresi Rad et al., 2024) learns bankfull width and depth from gage measurements and predicts them for all 2.65 million NHDPlus reaches in CONUS. Its training data come from HYDRoSWOT, which holds acoustic Doppler (ADCP) measurements through 2014. USGS now publishes every channel measurement it makes, by every method (wading, bridge, cableway, boat), through the Water Data OGC API: 4.77 million records. I rebuilt the training targets from those records using the RiverShapeML and AHGestimation conventions, kept the model, features and deployment pipeline unchanged, and asked one question: **does the added data improve the model where it is weakest?**

- **More training sites, mostly where they were scarcest.** Bankfull-width training sites rise from 4,375 to 6,928 (+58%). Sites under 10 km² rise from 66 to 111, and sites of 10–100 km² from 596 to 1,157.
- **The added measurements are sound.** Where non-ADCP and ADCP measurements were taken the same way at the same site and flow, widths agree within 0.1–2.4%.
- **Held-out error falls most in small streams.** For sites under 10 km², median absolute error in bankfull width falls from 74% to 41% and bias from +61% to +5% when the new sites are added. Overall error improves modestly (23.3% → 22.4%), and accuracy on the original HYDRoSWOT targets is unchanged.
- **A re-predicted national product with 90% prediction intervals.** The retrained model was run through the NOAA-OWP deployment pipeline for every reach, with split-conformal intervals by drainage-area class. It sits closer to the Bieger et al. (2015) regional curves in headwaters and removes most physically implausible values.

The work builds directly on Arash Modaresi Rad's RiverShapeML and the public NOAA-OWP/predict-riverML pipeline, and uses Mike Johnson's AHGestimation package for the target fits. The deployment pipeline is clean and complete enough that its national outputs reproduced exactly from the published models, which made a controlled experiment like this one possible in a few weeks.

1. Motivation

I walked a stretch of Shoal Creek in Austin, Texas, to see how a reach-averaged geometry product compares with the channel on the ground. Over a few hundred meters the channel moves from a walled, rock-lined section with a built low-flow channel, to an incised section held by gabions and a bedrock ledge, to a section rebuilding a small floodplain below a failed grade control. A single bankfull value for the whole 8.2 km reach (COMID 5781223) is the right abstraction for a national product, so that variability is not the subject here.

The gage at the downstream end of the walk is. USGS 08156675 (Shoal Creek at Silverway Drive) has **68 width measurements**. HYDRoSWOT has **2**, both boat measurements from 2009, so the site falls below the 5-measurement minimum. The other 66, mostly wading measurements, are not in

HYDRoSWOT at all. The question this note answers is how common that situation is, and what the model gains when those measurements are used.

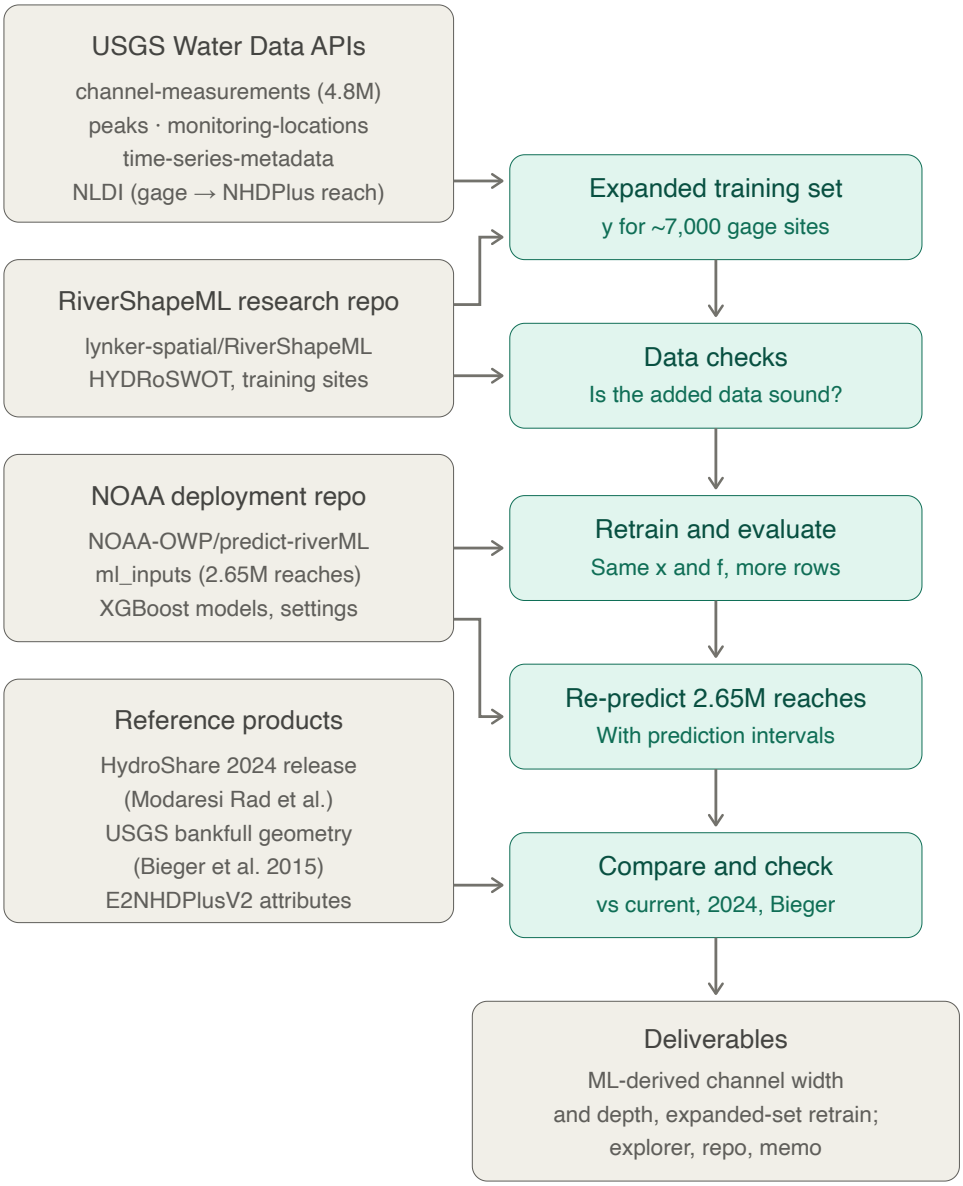


Figure 1. Process. Each box is a script in the accompanying repository, and the full table-level lineage (35 tables, 17 operations, each with source, retrieval date and commit) is checked automatically on every run.

2. Data and methods

Measurements. All USGS channel measurements were retrieved from the `channel-measurements` collection of the USGS Water Data OGC API (4,772,063 records), with site metadata and annual peaks from the `monitoring-locations` and `peaks` collections. Gages were linked to NHDPlus reaches with the NLDI; existing RiverShapeML sites keep their published COMIDs. Candidate new sites are CONUS gages

with at least 5 measurements since 2000 and at least 10 years of daily flow (4,501 sites; 3,192 linked to a reach).

Targets. For each site, width, mean depth (area / width) and velocity were converted to SI units and screened with RiverShapeML's sanity rules (width < 4 km, depth < 65 m, Q ≤ 1.1 × maximum annual peak). Hydraulic geometry was then fit with AHGestimation (`mad_filter` → `qva_filter` → `ahg_estimate`), which fits W = aQ^b, D = cQ^f and V = kQ^m jointly under continuity, and each curve was evaluated at the 2-year flood, Q_2. I took Q_2 as the median of the annual peak series, which is the 2-year flood by definition on an annual-maximum basis (England et al., 2019; Institute of Hydrology, 1999); it agrees with RiverShapeML's fitted values at shared sites (median ratio 0.97). The target is y = ln(1 + W_bf) with W_bf in meters, as in RiverShapeML, and likewise for depth. Sites where depth decreases with flow (f < 0) were dropped (53 sites).

Target filter. RiverShapeML keeps sites whose width–discharge fit has $R^2 \geq 0.6$. Many well-measured sites fail that test because width barely changes with flow, not because the target is uncertain: 3,636 of the 6,928 width targets here would fail it, with a median standard error of 0.037 on the log target. I replaced it with a precision filter: standard error of the log target at $Q_2 \leq 0.2$, $Q_2 \leq 3 \times$ the largest measured flow (limits extrapolation), and at least 5 measurements.

Model. Features, feature engineering, missing-value handling and inference follow predict-riverML exactly (30 features from `ml_inputs.parquet`). I first reproduced the deployed models' national predictions (width median 18.27 m vs 18.43 m in their saved statistics; depth identical). For retraining I kept their XGBoost learning rate and sampling settings but chose tree depth, minimum child weight and number of trees by a regularization sweep with early stopping on a reach-grouped 70/15/15 split (depth 6, min_child_weight 5, about 2,450 trees). This matched the deployed settings' held-out accuracy with roughly a quarter of the trees and much less fit to the training data (test log R^2 0.84 for width, 0.76 for depth). All runs below use the same settings, so only the training data differ.

Experiment. Three training sets, scored with 5-fold cross-validation grouped by reach, repeated with 3 seeds:

Run	Sites	Targets
A	Existing RiverShapeML sites	HYDRoSWOT measurements only
B1	Existing RiverShapeML sites	All USGS measurements
B2	B1 plus new sites	All USGS measurements

B1 versus B2 isolates the value of new sites. As a reproduction check, all three runs are also scored against HYDRoSWOT-only targets, to confirm that the added data do not degrade accuracy on the original problem. Error is reported as ln(predicted / observed), summarized as median absolute error (%), median bias (%), share of sites within a factor of 2, and R^2 in log space.

Prediction intervals. The B2 model was refit on all sites and applied to every reach through the deployment pipeline. Split-conformal 90% prediction intervals (Lei et al., 2018) were computed per drainage-area class from held-out residuals. Coverage was then checked on sites unseen by both the model and the calibration (0.88–0.90 across classes).

3. Results

More sites, concentrated in small drainages

Drainage area (km²)	Width sites today ($R^2 \geq 0.6$)	Expanded set: existing sites	Expanded set: new sites	Expanded total
< 10	66	38	73	111
10–100	596	560	597	1,157
100–1,000	1,766	1,919	876	2,795
1,000–10,000	1,395	1,620	465	2,085
> 10,000	485	577	193	770
Total	**4,375**	**4,719**	**2,209**	**6,928**

Totals include 10 sites without a drainage area. Depth targets follow the same pattern (7,017 sites). The gain is largest relative to need: 58% of NHDPlus reaches drain less than 10 km², and those reaches were represented by 66 training sites.

The added measurements are consistent with HYDRoSWOT

On a 300-site sample, 97% of HYDRoSWOT measurements matched a record in the USGS collection (same day, discharge within 3%), and a typical site has 13 HYDRoSWOT measurements against 142 in total. Where the same deployment type appears in both, widths agree closely (bridge +0.1%, boat +2.4%, cableway +1.0%). Wading measurements are about 33% narrower than bridge measurements at the same flows, because they are taken at different cross sections; this is a location difference rather than an error, and AHGestimation's filters and the joint fit absorb most of it. Repeat-measurement scatter sets a noise floor of 8–15% for width and 14–19% for depth. The joint AHG fit changes the median width target by less than 1% relative to a plain log-log fit but corrects poorly conditioned sites (e.g., Walnut Creek: 28.1 m → 19.0 m).

Held-out error falls most where data were added

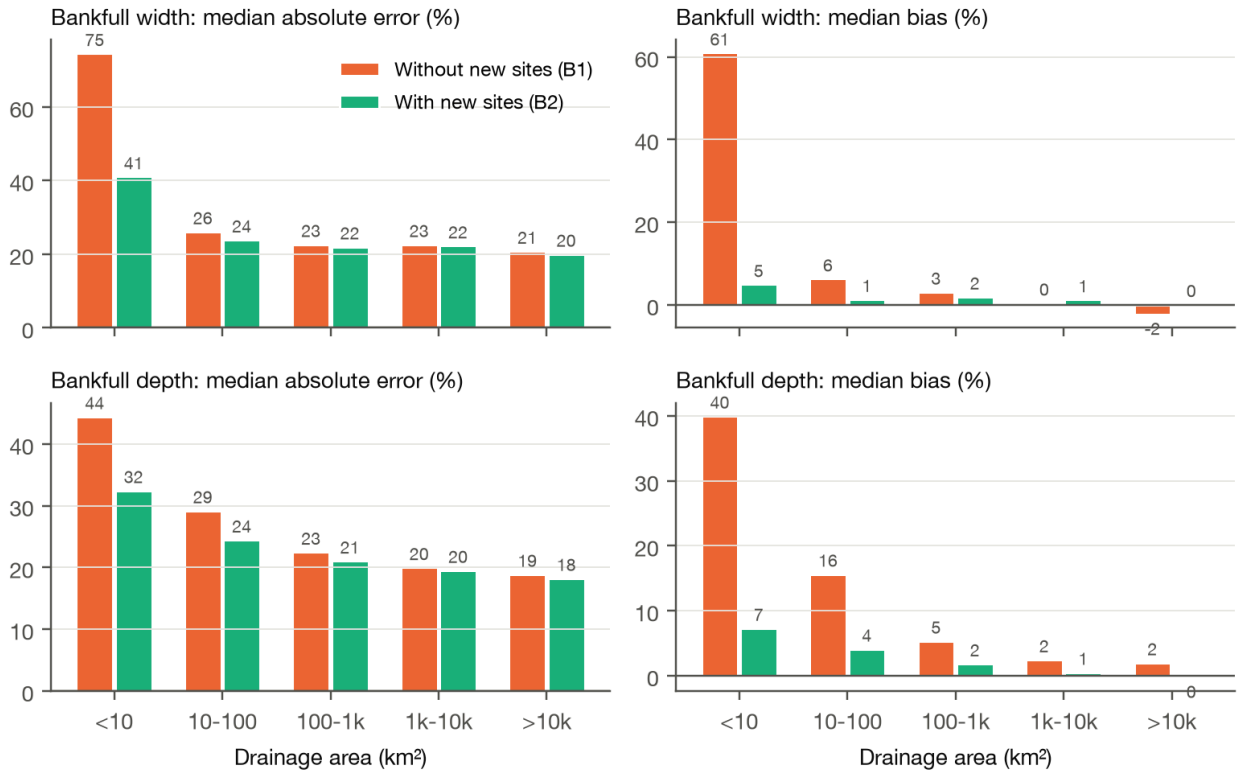


Figure 2. Held-out median absolute error and median bias by drainage-area class, without (B1) and with (B2) the new sites. Same model settings, same held-out sites.

Bankfull width	Median abs. error	Within 2×	Bias	Log R²
A (HYDRoSWOT targets)	25.4%	–	–	0.766
B1 (all measurements)	23.3%	90.5%	+2.4%	0.790
B2 (+ new sites)	22.4%	92.3%	+1.4%	0.816
B1 → B2, < 10 km²	74.5% → 41.1%	48% → 70%	+61% → +4.9%	0.39 → 0.64
B1 → B2, 10–100 km²	26.2% → 23.9%	–	+6.3% → +1.3%	0.43 → 0.56

For depth, overall error falls from 22.4% to 20.9% (log R^2 0.715 → 0.762), and for sites under 10 km² from 44.5% to 32.5% (bias +40% → +7%). In the largest basins (> 10,000 km²) the change is small (width 20.8% → 20.0%), as expected where data were already plentiful. For context, the repeat-measurement noise floor implies that no model can do much better than about 10–15% median error on width.

Reproduction check. Scored against the original HYDRoSWOT-only targets (3,658 sites), the three runs are indistinguishable: median absolute error 20.8% (A), 20.3% (B1), 20.0% (B2); log R^2 0.826, 0.813, 0.816. The added data help on the broader target set without costing accuracy on the original one.

A re-predicted national product

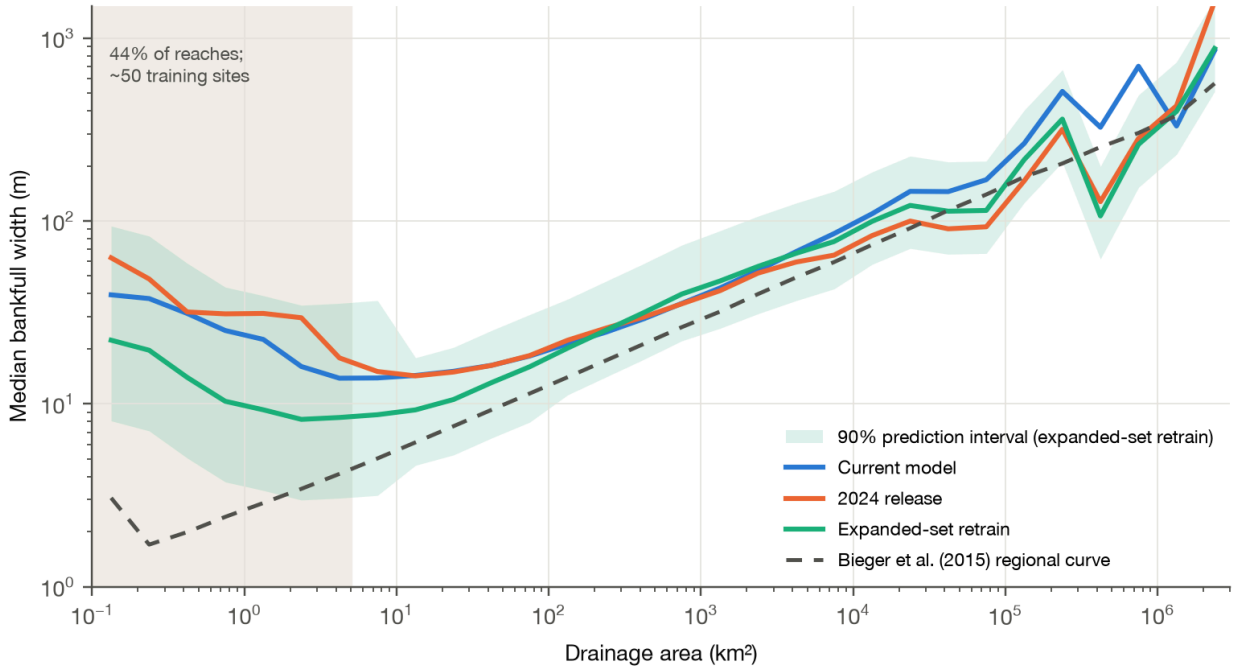


Figure 3. Median bankfull width by drainage area across all CONUS reaches for the current model (predictriverML), the 2024 HydroShare release, the expanded-set retrain with its 90% prediction interval, and the Bieger et al. (2015) regional curves. The shaded region holds 44% of reaches.

Check (all 2.6 M reaches)	Expanded-set retrain	Current model	2024 release
Width > 1 km	0.002%	0.47%	0.013%
Width / depth > 100	0.09%	1.94%	0.05%
Width halves going downstream	2.2%	3.6%	5.3%
Width–drainage-area exponent (Bieger: 0.36)	0.23	0.15	0.06
Depth–drainage-area exponent (Bieger: 0.25)	0.16	0.12	0.00
Median ratio to Bieger width	1.84	2.88	3.28

Above 100 km² the three ML products agree closely (median widths within about 10%). Below 10 km² they diverge: median widths are 9.1 m (expanded-set retrain), 17.6 m (current model), 26.5 m (2024 release) and 3.5 m (Bieger). Downstream scaling moves toward the regional curves, and implausibly wide reaches nearly disappear. Prediction intervals are honest about where the model knows least: for width they span roughly 0.55–1.9× the prediction above 10 km² and 0.36–4.2× below it.

The Silverway reach illustrates both the improvement and its limit. The retrain predicts 15.8 m wide and 0.99 m deep (current model 14.8 m / 1.29 m; 2024 release 19.6 m / 1.51 m). The Silverway site itself is not in the expanded training set: its Q_2 (2,700 cfs, 76 m³/s) is 5.2 times the largest measured flow, so its target would be an extrapolation and the filter excludes it.

4. Discussion and limitations

The headwater gap is narrower but not closed. Below about 5 km² there are still only ~50 training sites for 1.16 million reaches. Tree ensembles cannot extrapolate beyond the range of their training targets, so predictions flatten (and in all three products bend upward) at the smallest drainage areas rather than continuing the downstream trend. The ratio to Bieger widths remains 1.8 overall. Part of that gap is definitional: Bieger's field bankfull indicators and a Q_2-based target are not the same quantity, and the Q_2 convention likely overstates bankfull flow in small, flashy basins.

Gaged is not ungaged. All accuracy figures are for reaches with gages, which skew toward larger, more stable, more often modified channels. The conformal intervals inherit that assumption: they are calibrated on gaged sites and are exchangeable only to the extent that ungaged reaches resemble them.

Some useful data remain unused. About 1,300 candidate gages did not link to a reach through the NLDI, and about 7,000 sites have five or more measurements but less than 10 years of daily flow. The latter could use National Water Model flood frequencies, which the RiverShapeML codebase already supports; at gages NWM-based Q_2 agrees well overall (log R^2 ~0.80) but less well under 10 km² (0.47), which argues for using it with an uncertainty term.

Two observations I could not resolve from the public record. The current deployed models and the 2024 HydroShare release differ more than I expected (log R^2 ~0.35 for width between them), and the HydroShare file contains about 65,000 duplicated reach IDs. Both are probably explained by a model update between releases, and they are noted here as questions, not findings.

5. Questions and possible next steps

These are offered as starting points for a conversation; the team will know which matter and which have already been considered.

1. **Was the HYDRoSWOT-only choice deliberate?** For example, to keep a single, consistent measurement method. The agreement statistics above suggest the other methods can be added with little risk, but there may be reasons I have not seen.
2. **Headwater behavior.** Monotone constraints on drainage area and discharge, or blending with regional curves below a threshold, would keep predictions on a physically sensible trend where data are sparse.
3. **More sites.** Recover the unlinked gages and add NWM-based flood frequencies for the ~7,000 short-record sites.
4. **In-channel geometry.** The same measurements support targets at lower flows (in-channel flow is about 6% of bankfull flow at the median), and these sites are far better sampled there than at Q_2. This is the natural next experiment and maps directly onto the existing in-channel model.
5. **Training settings.** Early stopping and modest regularization reach the same accuracy with about a quarter of the trees, which shortens training and inference.
6. **Release housekeeping.** An updated HydroShare release could carry prediction intervals, and the duplicate IDs noted above.

I would be glad to prototype any of these within the predict-riverML structure so that the results drop straight into the existing pipeline.

Code and data

Repository: github.com/drmerlot/rivershapeml-more-data (private; access on request). It contains every script, a single `run_all.sh` that rebuilds all results from public sources, a checked data-lineage record, an interactive notebook (marimo) for exploring sites, targets, runs and the national product, and the re-predicted product ("ML-derived channel width and depth, expanded-set retrain") with 90% prediction intervals for every reach.

Acknowledgments

This work stands on Arash Modaresi Rad's RiverShapeML research and the NOAA Office of Water Prediction's open predict-riverML deployment pipeline, whose published models, inputs and inference code made exact reproduction straightforward. Target fitting uses AHGestimation by Mike Johnson, with contributions from Arash Modaresi Rad. Data are from the USGS (HYDRoSWOT; Water Data APIs; NLDI), the E2NHDPlus attributes, and the Bieger et al. (2015) regional curves (USGS ScienceBase). Any errors of interpretation are mine.

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